

13. Invariant Variational Problems

Nachr. v. d. Ges. d. Wiss. zu Göttingen 1918, pp. 235–257

Presented by F. Klein in the session of 26 July 1918.¹

The subject is variational problems that admit a continuous group (in Lie’s sense); the consequences which arise from this for the corresponding differential equations find their most general expression in the theorems formulated in §1 and proved in the following sections. For these differential equations arising from variational problems, much more precise assertions can be made than for arbitrary differential equations admitting a group, which form the subject of Lie’s investigations. What follows therefore rests on a combination of the methods of the formal calculus of variations with those of Lie group theory. For special groups and variational problems this combination of methods is not new; I mention Hamel and Herglotz for special finite groups, Lorentz and his students (for example Fokker), Weyl and Klein for special infinite groups.² In particular, the second note of Klein and the present developments have mutually influenced one another; for this I may refer to the closing remarks of Klein’s note.

§1. Preliminaries and formulation of the theorems

All functions occurring below are to be assumed analytic, or at least continuous and continuously differentiable a finite number of times, and single-valued in the domain under consideration.

By a “transformation group” one understands, as usual, a system of transformations such that for each transformation there exists an inverse transformation contained in the system, and such that the composition of any two transformations of the system again belongs to the system. The group is called a finite continuous $\mathfrak{G}_\rho$ when its transformations are contained in a most general transformation depending analytically on ρ essential parameters ε (that is, the ρ parameters are not to be representable as ρ functions of fewer parameters). Correspondingly, by an infinite continuous $\mathfrak{G}_{\infty\rho}$ one understands a group whose most general transformations depend on ρ essential arbitrary functions $p(x)$ and their derivatives, analytically or at least continuously and with a finite number of continuous derivatives. An intermediate case between the two is the group depending on infinitely many parameters but not on arbitrary functions. Finally, a mixed group denotes one that depends both on arbitrary functions and on parameters.³

Let $x_1, \dots, x_n$ be independent variables, and let $u_1(x), \dots, u_\mu(x)$ be functions depending on them. If one subjects the x ’s and u ’s to the transformations of a group, then, because the transformations are assumed invertible, among the transformed quantities there must again be exactly n independent ones, denoted $y_1, \dots, y_n$; the remaining quantities depending on these are denoted $v_1(y), \dots, v_\mu(y)$. The transformations may also contain the derivatives of the u ’s with respect to the x ’s, thus $\partial u/\partial x$, $\partial^2 u/\partial x^2, \dots$ ⁴ A function is called an invariant of the group if a relation holds:

$$P\left(x, u, \frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2}, \dots\right) = P\left(y, v, \frac{\partial v}{\partial y}, \frac{\partial^2 v}{\partial y^2}, \dots\right).$$

In particular, an integral I is therefore an invariant of the group if a relation holds:

$$\begin{aligned} I &= \int \cdots \int f\left(x, u, \frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2}, \dots\right) dx \\ &= \int \cdots \int f\left(y, v, \frac{\partial v}{\partial y}, \frac{\partial^2 v}{\partial y^2}, \dots\right) dy, \end{aligned} \tag{1}$$

¹The final version of the manuscript was submitted only at the end of September.

²Hamel: *Math. Ann.* vol. 59 and *Zeitschrift f. Math. u. Phys.* vol. 50. Herglotz: *Ann. d. Phys.* (4) vol. 36, especially §9, p. 511. Fokker, Verslag d. Amsterdamer Akad., 27 Jan. 1917. For the further literature compare the second note of Klein: *Göttinger Nachrichten*, 19 July 1918. In a paper by Kneser that has just appeared (*Math. Zeitschrift* vol. 2), the issue is the construction of invariants by a similar method.

³Lie defines, in *Grundlagen für die Theorie der unendlichen kontinuierlichen Transformationsgruppen* (Reports of the Royal Saxon Society of Sciences, 1891), the infinite continuous group as a transformation group whose transformations are given by the most general solutions of a system of partial differential equations, once these solutions do not depend only on a finite number of parameters. This gives one of the types indicated above, different from the finite group; conversely, however, the limiting case of infinitely many parameters need not necessarily satisfy a system of differential equations.

⁴I suppress the indices, as far as possible also in summations; thus, for example, $\partial^2 u/\partial x^2$ denotes $\partial^2 u_\nu/\partial x_i \partial x_k$, and so on.

where the integration is over an arbitrary real x -domain and the corresponding y -domain.⁵

On the other hand, for an arbitrary integral I , not necessarily invariant, I form the first variation δI , and transform it by partial integration according to the rules of the calculus of variations. As is well known, once δu , together with all its derivatives occurring at the boundary, is taken to vanish at the boundary, but is otherwise arbitrary, one obtains:

$$\delta I = \int \cdots \int \delta f \, dx = \int \cdots \int \left(\sum_i \psi_i \left(x, u, \frac{\partial u}{\partial x}, \dots \right) \delta u_i \right) dx, \quad (2)$$

where the ψ 's denote the Lagrange expressions, that is, the left-hand sides of the Lagrange equations of the corresponding variational problem $\delta I = 0$. This integral relation corresponds to an identity free of integrals in δu and its derivatives, which arises when one writes the boundary terms as well. As partial integration shows, these boundary terms are integrals over divergences, that is, over expressions

$$\text{Div } A = \frac{\partial A_1}{\partial x_1} + \cdots + \frac{\partial A_n}{\partial x_n},$$

where A is linear in δu and its derivatives. Thus one obtains:

$$\sum_i \psi_i \delta u_i = \delta f + \text{Div } A. \quad (3)$$

In particular, if f contains only first derivatives of the u 's, then, in the case of the simple integral, the identity (3) is identical with Herr's so-called "Lagrangian central equation":

$$\sum_i \psi_i \delta u_i = \delta f - \frac{d}{dx} \left(\sum_i \frac{\partial f}{\partial u'_i} \delta u_i \right), \quad \left(u'_i = \frac{du_i}{dx} \right), \quad (4)$$

whereas for the n -fold integral (3) becomes:

$$\sum_i \psi_i \delta u_i = \delta f - \frac{\partial}{\partial x_1} \left(\sum_i \frac{\partial f}{\partial \left(\frac{\partial u_i}{\partial x_1} \right)} \delta u_i \right) - \cdots - \frac{\partial}{\partial x_n} \left(\sum_i \frac{\partial f}{\partial \left(\frac{\partial u_i}{\partial x_n} \right)} \delta u_i \right). \quad (5)$$

For the simple integral and κ derivatives of the u 's, (3) is given by:

$$\begin{aligned} \sum_i \psi_i \delta u_i = & \delta f - \frac{d}{dx} \left\{ \sum_i \left[\binom{1}{1} \frac{\partial f}{\partial u_i^{(1)}} \delta u_i + \binom{2}{1} \frac{\partial f}{\partial u_i^{(2)}} \delta u_i^{(1)} + \cdots + \binom{\kappa}{1} \frac{\partial f}{\partial u_i^{(\kappa)}} \delta u_i^{(\kappa-1)} \right] \right\} \\ & + \frac{d^2}{dx^2} \left\{ \sum_i \left[\binom{2}{2} \frac{\partial f}{\partial u_i^{(2)}} \delta u_i + \binom{3}{2} \frac{\partial f}{\partial u_i^{(3)}} \delta u_i^{(1)} + \cdots + \binom{\kappa}{2} \frac{\partial f}{\partial u_i^{(\kappa)}} \delta u_i^{(\kappa-2)} \right] \right\} \\ & + \cdots + (-1)^\kappa \frac{d^\kappa}{dx^\kappa} \left\{ \sum_i \binom{\kappa}{\kappa} \frac{\partial f}{\partial u_i^{(\kappa)}} \delta u_i \right\}. \end{aligned} \quad (6)$$

A corresponding identity holds for the n -fold integral; in particular A contains δu up to the $(\kappa - 1)$ -st derivative. That (4), (5), and (6) do in fact define the Lagrange expressions ψ_i follows from the fact that, in the combinations on the right-hand sides, all higher derivatives of δu are eliminated, whereas the relation (2) is fulfilled, to which partial integration leads uniquely.

In what follows the issue is the two theorems:

I. If the integral I is invariant with respect to a $\mathfrak{G}_\rho$, then ρ linearly independent combinations of the Lagrange expressions become divergences; conversely, from this follows the invariance of I with respect to a $\mathfrak{G}_\rho$. The theorem also holds in the limiting case of infinitely many parameters.

II. If the integral I is invariant with respect to a $\mathfrak{G}_{\infty\rho}$ in which the arbitrary functions occur up to the σ -th derivative, then there exist ρ identical relations between the Lagrange expressions and their derivatives up to order σ ; here too the converse holds.⁶

⁵I write dx, dy , for short, for $dx_1 \cdots dx_n, dy_1 \cdots dy_n$. All arguments $x, u, \varepsilon, p(x)$ occurring in the transformations are to be taken as real, whereas the coefficients may be complex. But since the final results are identities in the x, u , the parameters, and the arbitrary functions, they also hold for complex values as soon as all occurring functions are assumed analytic. Moreover, a large part of the results can be founded without integrals, so that the restriction to the real is not necessary even for the justification. By contrast, the considerations at the end of §2 and the beginning of §5 do not seem to be feasible without integrals.

⁶For certain trivial exceptional cases compare §2, second note.

For mixed groups, the assertions of both theorems apply; hence there occur both dependencies and divergence relations independent of them.

Passing from these identities to the corresponding variational problem, and thus setting $\psi = 0$,⁷ Theorem I says in the one-dimensional case — where the divergence becomes a total differential — that ρ first integrals exist, although nonlinear dependencies may occur among them;⁸ in the multidimensional case one obtains the divergence equations that recently have often been called “conservation laws”. Theorem II says that ρ of the Lagrange equations are consequences of the remaining ones.

The simplest example of Theorem II — without the converse — is the Weierstrass parameter representation; here, under homogeneity of first order, the integral is known to be invariant if one replaces the independent variable x by an arbitrary function of x , while leaving u unchanged ($y = p(x)$; $v_i(y) = u_i(x)$). Thus one arbitrary function occurs, but without derivatives; and to this corresponds the well-known linear relation between the Lagrange expressions themselves:

$$\sum_i \psi_i \frac{du_i}{dx} = 0.$$

A further example is offered by the physicists’ “general theory of relativity”. Here one is dealing with the group of all transformations of the x ’s, $y_i = p_i(x)$, while the u ’s (denoted $g_{\mu\nu}$ and q) are subjected to the transformations thereby induced for the coefficients of a quadratic and a linear differential form, transformations that contain the first derivatives of the arbitrary functions $p(x)$. To this correspond the known n dependencies between the Lagrange expressions and their first derivatives.⁹

If, in particular, one specializes the group by admitting no derivatives of $u(x)$ in the transformations, and in addition by letting the transformed independent quantities depend only on the x ’s and not on the u ’s, then, as will be shown in §5, the invariance of I implies the relative invariance of $\sum_i \psi_i \delta u_i$,¹⁰ and likewise of the divergences occurring in Theorem I, as soon as the parameters are subjected to suitable transformations. It follows further that the first integrals mentioned above also admit the group. For Theorem II one similarly obtains the relative invariance of the left-hand sides of the dependencies, combined by means of the arbitrary functions; and, as a consequence, also a function whose divergence vanishes identically and which admits the group — the function that in the physicists’ theory of relativity mediates the connection between dependencies and energy theorem.¹¹ Finally, in group-theoretic form, Theorem II gives the proof of a related assertion of Hilbert concerning the failure of proper energy theorems in “general relativity”. With these supplementary remarks Theorem I contains all the theorems on first integrals known in mechanics and so forth, whereas Theorem II can be described as the greatest possible group-theoretic generalization of the “general theory of relativity”.

§2. Divergence relations and dependencies

Let $\mathfrak{G}$ be a continuous group, finite or infinite. It can always be arranged that the identical transformation corresponds to the value zero of the parameters ε , respectively of the arbitrary functions $p(x)$.¹² Thus the most general transformation will be of the form

$$y_i = A_i\left(x, u, \frac{\partial u}{\partial x}, \dots\right) = x_i + \Delta x_i + \dots, \quad v_i(y) = B_i\left(x, u, \frac{\partial u}{\partial x}, \dots\right) = u_i + \Delta u_i + \dots,$$

where $\Delta x_i, \Delta u_i$ denote the terms of lowest dimension in ε , respectively in $p(x)$ and its derivatives; they are to be assumed linear in these quantities. As will be shown later, this is no restriction of generality.

Now let the integral I be an invariant with respect to $\mathfrak{G}$, so that relation (1) is fulfilled. In particular, I is then also invariant with respect to the infinitesimal transformation contained in $\mathfrak{G}$:

$$y_i = x_i + \Delta x_i, \quad v_i(y) = u_i + \Delta u_i.$$

⁷Somewhat more generally one may also set $\psi_i = T_i$; compare §3, first note.

⁸Compare the end of §3.

⁹Compare, for example, Klein’s presentation.

¹⁰That is, $\sum_i \psi_i \delta u_i$ acquires a factor under transformations.

¹¹Compare the second note of Klein.

¹²Compare, for example, Lie, *Grundlagen*, p. 331. If arbitrary functions are involved, then the special values a^σ of the parameters are to be replaced by fixed functions p^σ , $\partial p^\sigma / \partial x, \dots$; and correspondingly the values $a^\sigma + \varepsilon$ by $p^\sigma + p(x)$, $\partial p^\sigma / \partial x + \partial p / \partial x$, and so on.

For this, relation (1) becomes:

$$0 = \Delta I = \int \cdots \int f\left(y, v(y), \frac{\partial v}{\partial y}, \dots\right) dy - \int \cdots \int f\left(x, u(x), \frac{\partial u}{\partial x}, \dots\right) dx. \quad (7)$$

The first integral is to be extended over the $x + \Delta x$ -domain corresponding to the x -domain. But this integration can also be transformed into an integration over the x -domain by means of the transformation valid for infinitesimal Δx :

$$\int \cdots \int f\left(y, v(y), \frac{\partial v}{\partial y}, \dots\right) dy = \int \cdots \int f\left(x, v(x), \frac{\partial v}{\partial x}, \dots\right) dx + \int \cdots \int \text{Div}(f \cdot \Delta x) dx. \quad (8)$$

If one therefore introduces, in place of the infinitesimal transformation Δu , the variation

$$\bar{\delta} u_i = v_i(x) - u_i(x) = \Delta u_i - \sum_{\lambda} \frac{\partial u_i}{\partial x_{\lambda}} \Delta x_{\lambda}, \quad (9)$$

then (7) and (8) become:

$$0 = \int \cdots \int \{\bar{\delta} f + \text{Div}(f \cdot \Delta x)\} dx. \quad (10)$$

Since the relation (10) is fulfilled when integrating over every arbitrary domain, the integrand must vanish identically. Thus the Lie differential equations for the invariance of I pass over into the relation

$$\bar{\delta} f + \text{Div}(f \cdot \Delta x) = 0. \quad (11)$$

If one expresses $\bar{\delta} f$ here, by (3), in terms of the Lagrange expressions, one obtains

$$\sum_i \psi_i \bar{\delta} u_i = \text{Div } B, \quad (B = A - f \cdot \Delta x), \quad (12)$$

and this relation therefore represents, for every invariant integral I , an identity in all occurring arguments; it is the desired form of the Lie differential equations for I .¹³

First assume $\mathfrak{G}$ to be a finite continuous group $\mathfrak{G}_{\rho}$. Since, by assumption, Δu and Δx are linear in the parameters $\varepsilon_1, \dots, \varepsilon_{\rho}$, by (9) the same holds for $\bar{\delta} u$ and its derivatives; hence A and B are linear in the ε . I therefore put

$$B = B^{(1)}\varepsilon_1 + \cdots + B^{(\rho)}\varepsilon_{\rho}, \quad \bar{\delta} u = \bar{\delta} u^{(1)}\varepsilon_1 + \cdots + \bar{\delta} u^{(\rho)}\varepsilon_{\rho},$$

where $\bar{\delta} u^{(1)}, \dots$ are functions of $x, u, \partial u / \partial x, \dots$. From (12) follow the desired divergence relations:

$$\sum_i \psi_i \bar{\delta} u_i^{(1)} = \text{Div } B^{(1)}, \dots, \quad \sum_i \psi_i \bar{\delta} u_i^{(\rho)} = \text{Div } B^{(\rho)}. \quad (13)$$

Thus ρ linearly independent combinations of the Lagrange expressions become divergences. The linear independence follows from the fact that, by (9), $\bar{\delta} u = 0, \Delta x = 0$ would imply $\Delta u = 0, \Delta x = 0$, and hence a dependency between the infinitesimal transformations. But by hypothesis no such dependency holds for any parameter value; otherwise the $\mathfrak{G}_{\rho}$ arising again by integration from the infinitesimal transformations would depend on fewer than ρ essential parameters. The further possibility $\bar{\delta} u = 0, \text{Div}(f \cdot \Delta x) = 0$ has been excluded. These conclusions remain valid in the limiting case of infinitely many parameters.

Now let $\mathfrak{G}$ be an infinite continuous group $\mathfrak{G}_{\infty\rho}$. Then again $\bar{\delta} u$ and its derivatives, hence also B , are linear in the arbitrary functions $p(x)$ and their derivatives.¹⁴ Substitution of the values of $\bar{\delta} u$, still independently of (12), gives:

$$\sum_i \psi_i \bar{\delta} u_i = \sum_{\lambda, i} \psi_i \left\{ a_i^{(\lambda)}(x, u, \dots) p^{(\lambda)}(x) + b_i^{(\lambda)}(x, u, \dots) \frac{\partial p^{(\lambda)}}{\partial x} + \cdots + c_i^{(\lambda)}(x, u, \dots) \frac{\partial^{\sigma} p^{(\lambda)}}{\partial x^{\sigma}} \right\}.$$

¹³(12) becomes $0 = 0$ for the trivial case — which can occur only if $\Delta x, \Delta u$ also depend on derivatives of u — when $\text{Div}(f \cdot \Delta x) = 0, \bar{\delta} u = 0$. These infinitesimal transformations must therefore always be split off from the groups, and in the formulation of the theorems only the number of the remaining parameters or arbitrary functions is to be counted. Whether the remaining infinitesimal transformations still always form a group must be left open.

¹⁴That it is no restriction to assume the p 's independent of $u, \partial u / \partial x, \dots$ is shown by the converse.

Now, analogously to the formula for partial integration, by means of the identity

$$\varphi(x, u, \dots) \frac{\partial^\tau p(x)}{\partial x^\tau} = (-1)^\tau \frac{\partial^\tau \varphi}{\partial x^\tau} p(x) \quad \text{mod divergences,}$$

the derivatives of p can be replaced by p itself and by divergences that are linear in p and its derivatives. Thus:

$$\sum_i \psi_i \bar{\delta} u_i = \sum_\lambda \left\{ a_i^{(\lambda)} \psi_i - \frac{\partial}{\partial x} (b_i^{(\lambda)} \psi_i) + \dots + (-1)^\sigma \frac{\partial^\sigma}{\partial x^\sigma} (c_i^{(\lambda)} \psi_i) \right\} p^{(\lambda)} + \text{Div } \Gamma, \quad (14)$$

with summation over i in the brace. In combination with (12) this gives:

$$\sum_\lambda \left\{ a_i^{(\lambda)} \psi_i - \frac{\partial}{\partial x} (b_i^{(\lambda)} \psi_i) + \dots + (-1)^\sigma \frac{\partial^\sigma}{\partial x^\sigma} (c_i^{(\lambda)} \psi_i) \right\} p^{(\lambda)} = \text{Div}(B - \Gamma). \quad (15)$$

I now form the n -fold integral over (15), extended over an arbitrary domain, and choose the $p(x)$'s so that they, together with all derivatives occurring in $B - \Gamma$, vanish at the boundary. Since the integral over a divergence reduces to a boundary integral, the integral over the left-hand side of (15) also vanishes for arbitrary $p(x)$'s that vanish at the boundary with sufficiently many derivatives. From the usual arguments it follows that the integrand vanishes for every $p(x)$, giving the ρ relations

$$\sum_i \left\{ a_i^{(\lambda)} \psi_i - \frac{\partial}{\partial x} (b_i^{(\lambda)} \psi_i) + \dots + (-1)^\sigma \frac{\partial^\sigma}{\partial x^\sigma} (c_i^{(\lambda)} \psi_i) \right\} = 0, \quad (\lambda = 1, 2, \dots, \rho). \quad (16)$$

These are the desired dependencies between the Lagrange expressions and their derivatives when I is invariant with respect to $\mathfrak{G}_{\infty\rho}$. The linear independence follows as above, since the converse leads back to (12); and since one can again conclude from the infinitesimal transformations to the finite ones, as is carried out more fully in §4. Accordingly, for a $\mathfrak{G}_{\infty\rho}$, ρ arbitrary transformations already occur in the infinitesimal transformations. From (15) and (16) it follows further that $\text{Div}(B - \Gamma) = 0$.

If, corresponding to a "mixed group", one takes Δx and Δu to be linear in the ε 's and the $p(x)$'s, then by setting once the $p(x)$'s and once the ε 's equal to zero, one sees that both the divergence relations (13) and the dependencies (16) exist.

§3. The converse in the case of the finite group

To prove the converse, the preceding considerations must first be run through, in essence, in reverse order. From the validity of (13), multiplying by the ε 's and adding gives the validity of (12); and by means of the identity (3) one obtains a relation

$$\bar{\delta} f + \text{Div}(A - B) = 0.$$

Thus, if one sets $\Delta x = \frac{1}{f}(A - B)$, one thereby arrives at (11); by integration this finally gives (7), $\Delta I = 0$, hence the invariance of I with respect to the infinitesimal transformation determined by $\Delta x, \Delta u$, where the Δu 's are determined by (9) from Δx and $\bar{\delta} u$, and $\Delta x, \Delta u$ become linear in the parameters. But $\Delta I = 0$ implies, in the known way, the invariance of I with respect to the finite transformations arising by integration of the simultaneous system:

$$\frac{dx_i}{dt} = \Delta x_i, \quad \frac{du_i}{dt} = \Delta u_i, \quad \left\{ \begin{array}{l} x_i = y_i, \\ u_i = v_i \end{array} \right\} \quad \text{for } t = 0. \quad (17)$$

These finite transformations contain ρ parameters $a_1, \dots, a_\rho$, namely the combinations $t\varepsilon_1, \dots, t\varepsilon_\rho$. From the assumption that there are to be ρ , and only ρ , linearly independent divergence relations (13), it follows further that the finite transformations, as soon as they do not contain the derivatives $\partial u / \partial x$, always form a group. In the contrary case, at least one infinitesimal transformation arising by Lie's bracket process would not be a linear combination of the remaining ρ ; and since I also admits this transformation, there would be more than ρ linearly independent divergence relations. Or else this infinitesimal transformation would be of the special form $\bar{\delta} u = 0$, $\text{Div}(f \Delta x) = 0$. But then Δx or Δu would depend on derivatives, contrary to the assumption. Whether this case can occur when derivatives appear in Δx or Δu must be left open; then all functions Δx for which $\text{Div}(f \Delta x) = 0$ must still be added to the Δx determined above in order to obtain the group property again, but by agreement the parameters thereby added are not to be counted. This proves the converse.

It also follows from this converse that Δx and Δu may indeed be assumed linear in the parameters. If Δu and Δx were forms of higher degree in ε , then, because of the linear independence of the power products of the ε 's, entirely corresponding relations (13), only in larger number, would follow; by the converse, these imply invariance of I with respect to a group whose infinitesimal transformations contain the parameters linearly. If this group is to contain exactly ρ parameters, then there must be linear dependencies among the divergence relations originally obtained from the terms of higher degree in ε .

It should also be noted that, in the case where Δx and Δu also contain derivatives of the u 's, the finite transformations may depend on infinitely many derivatives of the u 's; for in this case the integration of (17), when determining

$$\frac{d^2 x_i}{dt^2}, \quad \frac{d^2 u_i}{dt^2},$$

leads to

$$\Delta\left(\frac{\partial u}{\partial x_\lambda}\right) = \frac{\partial \Delta u}{\partial x_\lambda} - \sum_\nu \frac{\partial u}{\partial x_\nu} \frac{\partial \Delta x_\nu}{\partial x_\lambda},$$

so that in general the number of derivatives of u grows at every step. An example is:

$$f = \frac{1}{2}u'^2, \quad \psi = -u'', \quad \psi \cdot x = \frac{d}{dx}\{(u - u'x)\}, \quad \bar{\delta}u = x\varepsilon,$$

$$\Delta x = -\frac{2u}{u'}\varepsilon, \quad \Delta u = \left(x - \frac{2u}{u'}\right)\varepsilon.$$

Since the Lagrange expressions of a divergence vanish identically, the converse finally shows the following: if I admits a $\mathfrak{G}_\rho$, then every integral that differs from I only by a boundary integral, that is, by an integral over a divergence, also admits a $\mathfrak{G}_\rho$ with the same $\bar{\delta}u$'s, while its infinitesimal transformations will in general contain derivatives of the u 's. Thus, corresponding to the preceding example,

$$f^* = \frac{1}{2} \left\{ u'^2 - \frac{d}{dx} \left(\frac{u^2}{x} \right) \right\}$$

admits the infinitesimal transformation $\Delta u = x\varepsilon$, $\Delta x = 0$; whereas derivatives of u occur in the infinitesimal transformations corresponding to f .

If one passes to the variational problem, that is, if one sets $\psi_i = 0$,¹⁵ then (13) becomes the equations

$$\text{Div } B^{(1)} = 0, \dots, \quad \text{Div } B^{(\rho)} = 0,$$

which are often called “conservation laws”. In the one-dimensional case this gives:

$$B^{(1)} = \text{const.}, \dots, \quad B^{(\rho)} = \text{const.}.$$

Here the B 's contain at most $(2\kappa - 1)$ -st derivatives of u (by (6)), as soon as Δu and Δx contain no derivatives higher than the κ -th derivatives occurring in f . Since ψ contains, in general, 2κ -th derivatives,¹⁶ one thus has the existence of ρ first integrals. That nonlinear dependencies may exist among them is shown again by the above f . To the linearly independent $\Delta u = \varepsilon_1$, $\Delta x = \varepsilon_2$ correspond the linearly independent relations

$$u'' = \frac{d}{dx}u', \quad u''u' = \frac{1}{2} \frac{d}{dx}(u')^2,$$

whereas among the first integrals $u' = \text{const.}$, $u'^2 = \text{const.}$ there is a nonlinear dependency. This is the elementary case in which $\Delta u, \Delta x$ contain no derivatives of u .¹⁷

¹⁵ $\psi_i = 0$, or somewhat more generally $\psi_i = T_i$, where the T_i 's are newly adjoined functions, are called “field equations” in physics. In the case $\psi_i = T_i$, the identities (13) pass into the equations $\text{Div } B^{(\lambda)} = \sum_i T_i \bar{\delta}u_i^{(\lambda)}$, which in physics are also still called conservation laws.

¹⁶As soon as f is nonlinear in the κ -th derivatives.

¹⁷Otherwise one has additionally $u''u'^{\lambda-1} = \text{const.}$ for every λ , corresponding to

$$u''(u')^{\lambda-1} = \frac{1}{\lambda} \frac{d}{dx}(u')^\lambda.$$

§4. The converse in the case of the infinite group

First it is to be shown that the assumption of linearity of Δx and Δu is no restriction. Here this follows, even without the converse, from the fact that $\mathfrak{G}_{\infty\rho}$ depends formally on ρ , and only ρ , arbitrary functions. Namely, in the nonlinear case, when transformations are composed, the terms of lowest order add and the number of arbitrary functions would increase. Indeed, suppose for instance that

$$\begin{aligned} y &= A\left(x, u, \frac{\partial u}{\partial x}, \dots; p\right) \\ &= x + \sum a(x, u, \dots)p^\nu + b(x, u, \dots)p^{\nu-1}\frac{\partial p}{\partial x} + c p^{\nu-2}\left(\frac{\partial p}{\partial x}\right)^2 + \dots + d\left(\frac{\partial p}{\partial x}\right)^\nu + \dots, \end{aligned}$$

where

$$p^\nu = (p^{(1)})^{\nu_1} \dots (p^{(\rho)})^{\nu_\rho},$$

and correspondingly

$$v = B\left(x, u, \frac{\partial u}{\partial x}, \dots; p\right).$$

Then, by composition with

$$z = A\left(y, v, \frac{\partial v}{\partial y}, \dots; q\right),$$

the terms of lowest order give

$$z = x + \sum a(p^\nu + q^\nu) + b\left\{p^{\nu-1}\frac{\partial p}{\partial x} + q^{\nu-1}\frac{\partial q}{\partial x}\right\} + c\left\{p^{\nu-2}\left(\frac{\partial p}{\partial x}\right)^2 + q^{\nu-2}\left(\frac{\partial q}{\partial x}\right)^2\right\} + \dots.$$

If here some coefficient other than a and b is different from zero, so that a term

$$p^{\nu-\sigma}\left(\frac{\partial p}{\partial x}\right)^\sigma + q^{\nu-\sigma}\left(\frac{\partial q}{\partial x}\right)^\sigma$$

actually occurs for $\sigma > 1$, then it cannot be written as the differential quotient of a single function, nor as a product of powers of such a quotient. Thus the number of arbitrary functions has increased against the hypothesis. If all coefficients other than a and b vanish, then, depending on the values of the exponents $\nu_1, \dots, \nu_\rho$, the second term is the differential quotient of the first (as, for example, always for a $\mathfrak{G}_{\infty 1}$), so that linearity actually occurs; or else the number of arbitrary functions increases here as well. Hence, because of linearity in $p(x)$, the infinitesimal transformations satisfy a system of linear partial differential equations; and, since the group property is fulfilled, they form an “infinite group of infinitesimal transformations” in Lie’s sense (Grundlagen, §10).

The converse now results similarly to the case of the finite group. The existence of the dependencies (16), after multiplication by $p^{(\lambda)}(x)$ and addition, leads by means of the identical transformation (14) to

$$\sum_i \psi_i \bar{\delta} u_i = \text{Div } \Gamma.$$

From this, as in §3, one obtains the determination of Δx and Δu , and the invariance of I with respect to these infinitesimal transformations, which in fact depend linearly on ρ arbitrary functions and their derivatives up to order σ . That these infinitesimal transformations, when they contain no derivatives $\partial u/\partial x, \dots$, certainly form a group follows as in §3, since otherwise more arbitrary functions would occur by composition, whereas by assumption there are to be only ρ dependencies (16). Thus they form an “infinite group of infinitesimal transformations”. But such a group consists (Grundlagen, Theorem VII, p. 391) of the most general infinitesimal transformations of a certain thereby defined “infinite group $\mathfrak{G}$ of finite transformations” in Lie’s sense. Every finite transformation is thereby generated from infinitesimal ones (Grundlagen, §7),¹⁸ and therefore arises by integration of the simultaneous system:

$$\frac{dx_i}{dt} = \Delta x_i, \quad \frac{du_i}{dt} = \Delta u_i, \quad \left\{ \begin{array}{l} x_i = y_i, \\ u_i = v \end{array} \right\} \quad \text{for } t = 0.$$

¹⁸From this it follows in particular that the group $\mathfrak{G}$ generated from the infinitesimal transformations $\Delta x, \Delta u$ of a $\mathfrak{G}_{\infty\rho}$ leads back again to $\mathfrak{G}_{\infty\rho}$. For $\mathfrak{G}_{\infty\rho}$ contains no infinitesimal transformations depending on arbitrary functions and different from $\Delta x, \Delta u$, and cannot contain independent transformations depending on parameters either, since otherwise it would be a mixed group. But by the preceding argument the finite transformations are determined by the infinitesimal ones.

It may be necessary, however, to let the arbitrary $p(x)$'s still depend on t . Thus $\mathfrak{G}$ in fact depends on ρ arbitrary functions. In particular, if it is sufficient to take $p(x)$ free of t , then this dependency is analytic in the arbitrary functions $q(x) = t p(x)$.¹⁹ If derivatives $\partial u/\partial x, \dots$ occur, it may be necessary to adjoin infinitesimal transformations $\delta u = 0$, $\text{Div}(f \Delta x) = 0$ before drawing the same conclusions.

Following an example of Lie (Grundlagen, §7), I now indicate a rather general case in which one can get as far as explicit formulae, which also show that the derivatives of the arbitrary functions occur up to order σ ; in this case the converse is therefore complete. These are groups of infinitesimal transformations to which there corresponds the group of all transformations of the x 's and the transformations of the u 's thereby "induced"; that is, transformations of the u 's in which Δu , and hence u , depends only on the arbitrary functions occurring in Δx . It is still assumed that the derivatives $\partial u/\partial x, \dots$ do not occur in Δu . Thus:

$$\Delta x_i = p^{(i)}(x), \quad \Delta u_i = \sum_{\lambda=1}^n \left\{ a_i^{(\lambda)}(x, u) p^{(\lambda)} + b_i^{(\lambda)} \frac{\partial p^{(\lambda)}}{\partial x} + \dots + c_i^{(\lambda)} \frac{\partial^\sigma p^{(\lambda)}}{\partial x^\sigma} \right\}.$$

Since the infinitesimal transformation $\Delta x = p(x)$ generates every transformation $x = y + g(y)$ with arbitrary $g(y)$, one can in particular determine $p(x)$ as a function of t so that the one-parameter group is generated:

$$x_i = y_i + t g_i(y), \quad (18)$$

which for $t = 0$ becomes the identity and for $t = 1$ becomes the desired $x = y + g(y)$. Differentiation of (18) gives

$$\frac{dx_i}{dt} = g_i(y) = p^{(i)}(x, t), \quad (19)$$

where $p(x, t)$ is determined from $g(y)$ by inversion of (18); conversely, (18) arises from (19) by means of the side condition $x_i = y_i$ for $t = 0$, by which the integral is uniquely fixed. By means of (18), the x 's in Δu can be replaced by the integration constants y and by t ; the $g(y)$'s then occur exactly up to their σ -th derivatives, since in

$$\frac{\partial p}{\partial x} = \sum_k \frac{\partial g}{\partial y_k} \frac{\partial y_k}{\partial x}$$

the $\partial y/\partial x$ are expressed by $\partial x/\partial y$, and in general $\partial^\sigma p/\partial x^\sigma$ is replaced by its value in

$$\frac{\partial g}{\partial y}, \dots, \frac{\partial^\sigma g}{\partial y^\sigma}, \dots, \frac{\partial^\sigma x}{\partial y^\sigma}.$$

For the determination of the u 's one therefore obtains the system of equations

$$\frac{du_i}{dt} = F_i \left(g(y), \frac{\partial g}{\partial y}, \dots, \frac{\partial^\sigma g}{\partial y^\sigma}, u, t \right), \quad (u_i = v_i \text{ for } t = 0),$$

in which only t and u are variables, while $g(y), \dots$ belong to the coefficient domain; hence integration gives

$$u_i = v_i + B_i \left(v, g(y), \frac{\partial g}{\partial y}, \dots, \frac{\partial^\sigma g}{\partial y^\sigma}, t \right)_{t=1}.$$

Thus one obtains transformations that depend exactly on σ derivatives of the arbitrary functions. The identity is contained in this for $g(y) = 0$, by (18); and the group property follows from the fact that the procedure just described yields every transformation $x = y + g(y)$, whereby the induced transformation of the u 's is uniquely determined, so that the group $\mathfrak{G}$ is exhausted.

It also follows incidentally from the converse that it is no restriction to assume that the arbitrary functions depend only on the x 's, and not on $u, \partial u/\partial x, \dots$. In the latter case, in the identical transformation (14), and hence also in (15), besides the $p^{(\lambda)}$'s there would occur the derivatives $\partial p^{(\lambda)}/\partial u$, $\partial p^{(\lambda)}/\partial(\partial u/\partial x), \dots$. If one now successively takes the $p^{(\lambda)}$'s to be of zeroth, first, and higher degrees in $u, \partial u/\partial x, \dots$, with arbitrary functions of x as coefficients, then dependencies (16) again arise, only in greater number. But by the preceding converse they can be brought back to the earlier case by being combined with arbitrary functions depending only on x . Similarly one shows that mixed groups correspond to the simultaneous occurrence of dependencies and independent divergence relations.²⁰

¹⁹The question whether this latter case perhaps always occurs was raised by Lie in another formulation (Grundlagen, §7 and §13, end).

²⁰As in §3, it follows here too from the converse that besides I , every integral I^* differing from it by an integral over a divergence also

§5. Invariance of the individual components of the relations

If the group $\mathfrak{G}$ is specialized to the simplest case, the one ordinarily considered, by admitting no derivatives of the u 's in the transformations and by letting the transformed independent variables depend only on the x 's and not on the u 's, then one can infer the invariance of the individual components in the formulae. First, by familiar arguments, one obtains the invariance of

$$\int \cdots \int \left(\sum_i \psi_i \delta u_i \right) dx,$$

hence the relative invariance of $\sum_i \psi_i \delta u_i$, where δ denotes any variation. Namely, on the one hand,

$$\delta I = \int \cdots \int \delta f \left(x, u, \frac{\partial u}{\partial x}, \dots \right) dx = \int \cdots \int \delta f \left(y, v, \frac{\partial v}{\partial y}, \dots \right) dy.$$

On the other hand, for $\delta u, \delta(\partial u/\partial x), \dots$ vanishing at the boundary, to which, because of the linear homogeneous transformation of $\delta u, \delta(\partial u/\partial x), \dots$, there corresponds $\delta v, \delta(\partial v/\partial y), \dots$ also vanishing at the boundary, one has

$$\begin{aligned} \int \cdots \int \delta f \left(x, u, \frac{\partial u}{\partial x}, \dots \right) dx &= \int \cdots \int \left(\sum_i \psi_i(u, \dots) \delta u_i \right) dx, \\ \int \cdots \int \delta f \left(y, v, \frac{\partial v}{\partial y}, \dots \right) dy &= \int \cdots \int \left(\sum_i \psi_i(v, \dots) \delta v_i \right) dy. \end{aligned}$$

Thus, for $\delta u, \delta(\partial u/\partial x), \dots$ vanishing at the boundary,

$$\begin{aligned} \int \cdots \int \left(\sum_i \psi_i(u, \dots) \delta u_i \right) dx &= \int \cdots \int \left(\sum_i \psi_i(v, \dots) \delta v_i \right) dy \\ &= \int \cdots \int \left(\sum_i \psi_i(v, \dots) \delta v_i \right) \left| \frac{\partial y_i}{\partial x_k} \right| dx. \end{aligned}$$

If one expresses $y, v, \delta v$ in the third integral by $x, u, \delta u$, and sets it equal to the first, then one obtains a relation

$$\int \cdots \int \left(\sum_i \chi_i(u, \dots) \delta u_i \right) dx = 0$$

for δu vanishing at the boundary but otherwise arbitrary. From this it follows, as is known, that the integrand vanishes for arbitrary δu ; hence the relation

$$\sum_i \psi_i(u, \dots) \delta u_i = \left| \frac{\partial y_i}{\partial x_k} \right| \left(\sum_i \psi_i(v, \dots) \delta v_i \right)$$

admits an infinite group with the same δu 's; here, however, Δx and Δu will in general contain derivatives of the u 's. Einstein introduced such an integral I^* into general relativity in order to obtain a simpler formulation of the energy theorems. I give the infinitesimal transformations admitted by this I^* , using exactly the notation of Klein's second note. The integral $I = \int \cdots \int K d\omega = \int \cdots \int \mathfrak{R} dS$ admits the group of all transformations of the w 's and the group thereby induced for the $g_{\mu\nu}$'s. To this correspond the dependencies ((30) in Klein):

$$\sum \mathfrak{R}_{\mu\nu} g_{\mu\nu}^{\alpha\beta} + 2 \sum \frac{\partial g_{\mu\nu}^{\alpha\beta} \mathfrak{R}_{\mu\nu}}{\partial w^\sigma} = 0.$$

Now $I^* = \int \cdots \int \mathfrak{R}^* dS$, where $\mathfrak{R}^* = \mathfrak{R} + \text{Div}$, and consequently $\mathfrak{R}_{\mu\nu}^* = \mathfrak{R}_{\mu\nu}$, where $\mathfrak{R}_{\mu\nu}^*, \mathfrak{R}_{\mu\nu}$ denote the Lagrange expressions respectively. The given dependencies are therefore also dependencies for $\mathfrak{R}_{\mu\nu}^*$; and after multiplication by p^τ and addition, by reversing the transformations of product differentiation, one obtains

$$\begin{aligned} \sum \mathfrak{R}_{\mu\nu} p^{\mu\nu} + 2 \text{Div} \left(\sum g^{\mu\sigma} \mathfrak{R}_{\mu\tau} p^\tau \right) &= 0, \\ \delta \mathfrak{R}^* + \text{Div} \left(\sum 2g^{\mu\sigma} \mathfrak{R}_{\mu\tau} p^\tau - \frac{\partial \mathfrak{R}^*}{\partial g_{\mu\nu}^\sigma} p^{\mu\nu} \right) &= 0. \end{aligned}$$

Comparison with the Lie differential equation $\delta \mathfrak{R}^* + \text{Div}(\mathfrak{R}^* \Delta w) = 0$ gives

$$\Delta w^\sigma = \frac{1}{\mathfrak{R}^*} \left(\sum 2g^{\mu\sigma} \mathfrak{R}_{\mu\tau} p^\tau - \frac{\partial \mathfrak{R}^*}{\partial g_{\mu\nu}^\sigma} p^{\mu\nu} \right), \quad \Delta g^{\mu\nu} = p^{\mu\nu} + \sum g_{\sigma}^{\mu\nu} \Delta w^\sigma,$$

as infinitesimal transformations admitted by I^* . These infinitesimal transformations therefore depend on the first and second derivatives of the $g^{\mu\nu}$'s, and contain the arbitrary p 's up to the first derivative.

holds identically in δu . This expresses the relative invariance of $\sum_i \psi_i \delta u_i$, and consequently the invariance of

$$\int \cdots \int \left(\sum_i \psi_i \delta u_i \right) dx.$$

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To apply this to the derived divergence relations and dependencies, one must first show that the $\bar{\delta}u$ derived from $\Delta u, \Delta x$ actually satisfies the transformation laws for the variation δu , provided only that the parameters, respectively arbitrary functions, in $\bar{\delta}v$ are determined as they correspond to the similar group of infinitesimal transformations in y, v . Let T_q denote the transformation that carries x, u into y, v ; and let T_p be an infinitesimal transformation in x, u . Then the similar one in y, v is given by $T_r = T_q T_p T_q^{-1}$, where the parameters, respectively arbitrary functions, r are determined by p and q . In formulae this is expressed as follows:

$$\begin{aligned} T_p : \quad \xi &= x + \Delta x(x, p), & u^* &= u + \Delta u(x, u, p); \\ T_q : \quad y &= A(x, q), & v &= B(x, u, q); \\ T_q T_p : \quad \eta &= A(x + \Delta x(x, p), q), & v^* &= B(x + \Delta x(p), u + \Delta u(p), q). \end{aligned}$$

But from this arises $T_r = T_q T_p T_q^{-1}$, hence

$$\eta = y + \Delta y(r), \quad v^* = v + \Delta v(r),$$

where, by means of the inverse of T_q , one regards the x 's as functions of the y 's and retains only the infinitesimal terms. Thus one has the identity

$$\eta = y + \Delta y(r) = y + \sum \frac{\partial A(x, q)}{\partial x} \Delta x(p), \quad (20a)$$

$$v^* = v + \Delta v(r) = v + \sum \frac{\partial B(x, u, q)}{\partial x} \Delta x(p) + \sum \frac{\partial B(x, u, q)}{\partial u} \Delta u(p). \quad (20b)$$

If, in this, one replaces $\xi = x + \Delta x$ by $\xi - \Delta \xi$, so that ξ goes back into x and hence Δx vanishes, then by the first formula (20) η also goes back into $y = \eta - \Delta \eta$. If under this substitution $\Delta u(p)$ goes over into $\bar{\delta}u(p)$, then $\Delta v(r)$ also goes over into $\bar{\delta}v(r)$, and the second formula (20) gives

$$\begin{aligned} v + \bar{\delta}v(y, v, \dots, r) &= v + \sum \frac{\partial B(x, u, q)}{\partial u} \bar{\delta}u(p), \\ \bar{\delta}v(y, v, \dots, r) &= \sum \frac{\partial B}{\partial u_\kappa} \bar{\delta}u_\kappa(x, u, p). \end{aligned}$$

Thus the transformation formulae for variations are indeed satisfied, provided only that $\bar{\delta}v$ is taken to depend on the parameters, respectively arbitrary functions, r .²²

It follows in particular that $\sum_i \psi_i \bar{\delta}u_i$ is relatively invariant; hence also, by (12), since the divergence relations are also fulfilled in y, v , the relative invariance of $\text{Div } B$; and further, by (14) and (13), the relative invariance of $\text{Div } \Gamma$ and of the left-hand sides of the dependencies combined with the $p^{(\lambda)}$'s, where in the transformed formulae the arbitrary $p(x)$'s (respectively the parameters) are everywhere to be replaced by the r 's. From this follows also the relative invariance of $\text{Div}(B - \Gamma)$, that is, of the divergence of a system of functions $B - \Gamma$ that does not vanish identically, but whose divergence vanishes identically.

From the relative invariance of $\text{Div } B$, in the one-dimensional case and for a finite group, one can draw a further conclusion about the invariance of the first integrals. The transformation of the parameters corresponding to the infinitesimal transformation becomes, by (20), linear and homogeneous; and because all transformations are invertible, the ε 's are also linear and homogeneous in the transformed parameters

²¹These arguments fail if y also depends on the u 's, because then $\delta f(y, v, \partial v / \partial y, \dots)$ also contains terms $\sum (\partial f / \partial y) \delta y$, so that the divergence transformation does not lead to the Lagrange expressions. They likewise fail if derivatives of the u 's are admitted; then the δv 's become linear combinations of $\delta u, \delta(\partial u / \partial x), \dots$, and hence only after a further divergence transformation lead to an identity $\int \cdots \int (\sum \chi_i(u, \dots) \delta u_i) dx = 0$, so that on the right the Lagrange expressions again do not occur. The question whether the invariance of $\int \cdots \int (\sum \psi_i \delta u_i) dx$ already allows one to conclude the existence of divergence relations is, by the converse, equivalent to the question whether it implies invariance of I with respect to a group that need not lead to the same $\Delta u, \Delta x$, but does lead to the same $\bar{\delta}u$. In the special case of the simple integral and only first derivatives in f , for the finite group one can infer the existence of first integrals from the invariance of the Lagrange expressions (compare, for example, Engel, Gött. Nachr. 1916, p. 270).

²²It is again seen that y must be assumed independent of u , and so on, if the conclusions are to hold. As an example, one may cite the $\delta g^{\mu\nu}$ and δq_σ indicated by Klein, which satisfy the transformations for variations as soon as the p 's are subjected to a vector transformation.

ε^* . This invertibility is certainly preserved when one sets $\psi = 0$, since no derivatives of u occur in (20). By equating the coefficients of the ε^* in

$$\text{Div } B(x, u, \dots, \varepsilon) = \frac{dy}{dx} \text{Div } B(y, v, \dots, \varepsilon^*)$$

the quantities $\frac{d}{dy} B^{(\lambda)}(y, v, \dots)$ also become linear homogeneous functions of the $\frac{d}{dx} B^{(\lambda)}(x, u, \dots)$. Thus from

$$\frac{d}{dx} B^{(\lambda)}(x, u, \dots) = 0 \quad \text{or} \quad B^{(\lambda)}(x, u) = \text{const.}$$

it follows that

$$\frac{d}{dy} B^{(\lambda)}(y, v, \dots) = 0 \quad \text{or} \quad B^{(\lambda)}(y, v) = \text{const.}$$

The ρ first integrals corresponding to a $\mathfrak{G}_\rho$ therefore each admit the group, so that the further integration is also simplified. The simplest example of this is that f is free of x or of a u , corresponding to the infinitesimal transformations $\Delta x = \varepsilon, \Delta u = 0$, respectively $\Delta x = 0, \Delta u = \varepsilon$. Then $\bar{\delta}u = -\varepsilon du/dx$, respectively ε ; and since B is derived from f and $\bar{\delta}u$ by differentiation and rational operations, it is also free of x , respectively u , and admits the corresponding groups.²³

§6. A theorem of Hilbert

From the foregoing there finally results the proof of an assertion of Hilbert concerning the connection between the failure of proper energy theorems and “general relativity” (Klein’s first note, Göttinger Nachr. 1917, Reply, first paragraph), in a generalized group-theoretic formulation.

Suppose the integral I admits a $\mathfrak{G}_{\infty\rho}$, and let $\mathfrak{G}_\sigma$ be any finite group arising by specialization of the arbitrary functions, thus a subgroup of $\mathfrak{G}_{\infty\rho}$. To the infinite group $\mathfrak{G}_{\infty\rho}$ there correspond dependencies (16), to the finite group $\mathfrak{G}_\sigma$ divergence relations (13); conversely, from the existence of any divergence relations follows the invariance of I with respect to a finite group, which is identical with $\mathfrak{G}_\sigma$ if and only if the $\bar{\delta}u$ ’s are linear combinations of those arising from $\mathfrak{G}_\sigma$. Hence the invariance with respect to $\mathfrak{G}_\sigma$ can lead to no divergence relations different from (13). But since from the existence of (16) follows the invariance of I with respect to the infinitesimal transformations $\Delta u, \Delta x$ of $\mathfrak{G}_{\infty\rho}$ for arbitrary $p(x)$, it follows in particular that I is already invariant with respect to the infinitesimal transformations of $\mathfrak{G}_\sigma$ arising from these by specialization, and hence with respect to $\mathfrak{G}_\sigma$. The divergence relations

$$\sum_i \psi_i \bar{\delta} u_i^{(\lambda)} = \text{Div } B^{(\lambda)}$$

must therefore be consequences of the dependencies (16), which can also be written

$$\sum_i \psi_i a_i^{(\lambda)} = \text{Div } \chi^{(\lambda)},$$

where the $\chi^{(\lambda)}$ ’s are linear combinations of the Lagrange expressions and their derivatives. Since the ψ ’s enter linearly both in (13) and in (16), the divergence relations must in particular be linear combinations of the dependencies (16). Thus

$$\text{Div } B^{(\lambda)} = \text{Div} \left(\sum \alpha_\mu \chi^{(\mu)} \right),$$

and the $B^{(\lambda)}$ ’s themselves are thus composed linearly from the χ ’s, that is, from the Lagrange expressions and their derivatives, and from functions whose divergence vanishes identically, as for instance the $B - \Gamma$ occurring at the end of §2, for which $\text{Div}(B - \Gamma) = 0$, and where the divergence has at the same time the character of an invariant. I shall call divergence relations in which the $B^{(\lambda)}$ ’s can be composed in the indicated way from the Lagrange expressions and their derivatives “improper”, and all the remaining ones “proper”.

²³In those cases where the existence of first integrals already follows from the invariance of $\int (\sum \psi_i \delta u_i) dx$, these do not admit the complete group $\mathfrak{G}_\rho$. For example, $\int (u'' \delta u) dx$ admits the infinitesimal transformation $\Delta x = \varepsilon_2, \Delta u = \varepsilon_1 + x \varepsilon_3$; whereas the first integral $u - u'x = \text{const.}$, corresponding to $\Delta x = 0, \Delta u = x \varepsilon_3$, does not admit the two other infinitesimal transformations, since it contains both u and x explicitly. To this first integral there correspond infinitesimal transformations for f that contain derivatives. Thus one sees that the invariance of $\int \dots \int (\sum \psi_i \delta u_i) dx$ in any case accomplishes less than the invariance of I , a point relevant to the question raised in a preceding note.

Conversely, if the divergence relations are linear combinations of the dependencies (16), hence “improper”, then the invariance with respect to $\mathfrak{G}_\sigma$ follows from that with respect to $\mathfrak{G}_{\infty\rho}$; $\mathfrak{G}_\sigma$ becomes a subgroup of $\mathfrak{G}_{\infty\rho}$. Thus the divergence relations corresponding to a finite group $\mathfrak{G}_\sigma$ become improper if and only if $\mathfrak{G}_\sigma$ is a subgroup of an infinite group with respect to which I is invariant.

By specialization of the groups, Hilbert’s original assertion follows from this. By the “translation group” one means the finite group

$$y_i = x_i + \varepsilon_i, \quad v_i(y) = u_i(x);$$

hence

$$\Delta x_i = \varepsilon_i, \quad \Delta u_i = 0, \quad \bar{\delta} u_i = - \sum_{\lambda} \frac{\partial u_i}{\partial x_{\lambda}} \varepsilon_{\lambda}.$$

Invariance with respect to the translation group says, as is known, that in

$$I = \int \cdots \int f\left(x, u, \frac{\partial u}{\partial x}, \dots\right) dx$$

the x ’s do not occur explicitly in f . The associated n divergence relations

$$\sum_i \psi_i \frac{\partial u_i}{\partial x_{\lambda}} = \text{Div } B^{(\lambda)} \quad (\lambda = 1, 2, \dots, n)$$

are to be called “energy relations”, since the “conservation laws” $\text{Div } B^{(\lambda)} = 0$ corresponding to the variational problem correspond to the “energy theorems”, and the $B^{(\lambda)}$ ’s to the “energy components”. Thus one obtains: if I admits the translation group, then the energy relations become improper if and only if I is invariant with respect to an infinite group containing the translation group as a subgroup.²⁴

An example of such infinite groups is given by the group of all transformations of the x ’s and of the induced transformations of the $u(x)$ ’s in which only derivatives of the arbitrary functions $p(x)$ occur. The translation group arises by the specialization $p^{(i)}(x) = \varepsilon_i$. It must remain undecided, however, whether by this means — and by the groups arising through changing I by a boundary integral — the most general such groups have already been given. Induced transformations of the indicated kind arise, for example, by subjecting the u ’s to the coefficient transformations of a “total differential form”, that is, of a form

$$\sum a \, d^2 x_i + \sum b \, dx_i dx_k + \cdots,$$

which contains, besides the dx ’s, also higher differentials. More special induced transformations, in which the $p(x)$ ’s occur only in first derivatives, are given by the coefficient transformations of ordinary differential forms $\sum c \, dx_{i_1} \cdots dx_{i_{\lambda}}$; these are the ones that have usually been considered.

A further group of the indicated kind — which, because of the occurrence of the logarithmic term, cannot be a coefficient transformation — is, for example, the following:

$$y = x + p(x), \quad v_i = u_i + \lg(1 + p'(x)) = u_i + \lg \frac{dy}{dx};$$

$$\Delta x = p(x), \quad \Delta u_i = p'(x), \quad \bar{\delta} u_i = p'(x) - u'_i p(x).$$

Here the dependencies (16) become

$$\sum_i \left(\psi_i u'_i + \frac{d\psi_i}{dx} \right) = 0,$$

and the improper energy relations become

$$\sum_i \left(\psi_i u'_i + \frac{d(\psi_i + \text{const.})}{dx} \right) = 0.$$

A simplest invariant integral of the group is

$$I = \int \frac{e^{-2u_1}}{u'_1 - u'_2} dx.$$

²⁴The energy theorems of classical mechanics, and likewise those of the old “theory of relativity” (where $\sum dx^2$ goes over into itself), are “proper”, since no infinite groups occur here.

The most general I is determined by integration of the Lie differential equation (11)

$$\bar{\delta}f + \frac{d}{dx}(f \cdot \Delta x) = 0,$$

which, after substituting the values for Δx and $\bar{\delta}u$, becomes, as soon as one assumes f to depend on only first derivatives of the u 's,

$$\frac{\partial f}{\partial x}p(x) + \left\{ \sum_i \frac{\partial f}{\partial u_i} - \sum_i \frac{\partial f}{\partial u'_i} u'_i + f \right\} p'(x) + \left\{ \sum_i \frac{\partial f}{\partial u'_i} \right\} p''(x) = 0$$

(identically in $p(x), p'(x), p''(x)$). Already for two functions $u(x)$, this system of equations has solutions that actually contain the derivatives, namely

$$f = (u'_1 - u'_2) \Phi \left(u_1 - u_2, \frac{e^{-u_1}}{u'_1 - u'_2} \right),$$

where Φ denotes an arbitrary function of the indicated arguments.

Hilbert states his assertion by saying that the failure of proper energy theorems is a characteristic feature of the “general theory of relativity”. For this assertion to hold literally, the designation “general relativity” must therefore be understood more broadly than usual, and extended also to the preceding groups depending on n arbitrary functions.²⁵

²⁵This again confirms the correctness of a remark of Klein, that the term “relativity” customary in physics should be replaced by “invariance relative to a group”. (Über die geometrischen Grundlagen der Lorentzgruppe. *Jhrber. d. d. Math. Vereinig.* vol. 19, p. 287, 1910; reprinted in the *Physikalische Zeitschrift*.)